

Building an End-to-End Test Automation Framework

Project Scope and Objectives

Problem Definition

Test automation framework that ensures:

- Functional correctness
- API reliability
- Business flow validation using BDD

Project Goals (SMART)

- Automate **80% of regression test cases**
- Achieve **<5% test failure due to flakiness**
- Cover **UI + API + End-to-End flows**
- Integrate reporting (Allure)
- Complete within **7 days capstone timeline**

Literature Review

- Selenium for UI automation
- Behave for BDD (Gherkin syntax)
- Postman/Newman for API automation
- Page Object Model (POM) for maintainability

Identified Gaps

- Poor integration between UI + API tests
- Hardcoded test data
- Lack of reusable step definitions
- Weak reporting & traceability

Proposal

- Combine **UI + API + BDD**
- Use **data-driven + modular framework**
- Provide **CI-ready automation**

Formulating Testing Hypothesis

Hypothesis

- **H₀ (Null):** Automated testing does not significantly improve defect detection rate.
- **H_a (Alternate):** Automated testing significantly improves defect detection and reduces regression time.

Test Data & Environment Setup

Test Data Acquisition

- User credentials (valid/invalid)
- Product catalog data
- Order scenarios
- API payloads (JSON)

Data Preparation

- Store test data in:
 - JSON / CSV / Excel
- Use dynamic data generation (faker library)
- Environment configs:
 - Dev / QA / Staging URLs

Test Analysis

Exploratory Testing

- Identify critical workflows:
 - Login → Search → Add to Cart → Checkout
- Identify edge cases:
 - Invalid inputs
 - Session timeout
 - API failure scenarios

Insights

- Checkout is the most failure-prone module
- APIs fail under invalid payloads
- UI inconsistencies across browsers

Framework Design

UI Automation (Selenium + Python)

- Page Object Model (POM)
- Reusable utilities
- Explicit waits

BDD Layer (Behave)

- Feature files (Gherkin)
- Step definitions
- Hooks (before/after)

API Testing (Postman)

- Collections
- Environment variables
- Post request scripts
- Tests using JS

Architecture

project

- tests/
- features/
- steps/
- pages/
- utils/
- data/
- reports/

Execution Phase

Test Execution

- Run Selenium scripts
- Execute Behave scenarios
- Run Postman collections via Newman

Evaluation Metrics

- Pass/Fail %
- Defect Density
- Test Coverage
- Execution Time
- Flaky Test Rate

Results & Analysis

Visualizations

- Test execution reports (Allure)
- Module-wise defects

Key Findings

- Automation reduced regression time by **70%**
- API failures detected earlier than UI
- BDD improved communication with stakeholders

Documentation & Reporting

Artifacts

- Test Cases
- Automation Scripts
- API Collections

Report

- Framework design
- Test coverage
- Execution results
- Defect summary

Presentation

- Demo
 - UI automation
 - API testing
 - BDD scenarios
- Show reports & insights

Ethical Considerations

Testing Ethics

- No real user data used
- Secure handling of credentials
- Avoid testing on production systems

Project Management

Timeline (7 Days)

Day	Task
Day 1	Requirement / Feature / Module Analysis
Day 2	Framework Setup
Day 3,4	Selenium Automation
Day 5	BDD Implementation
Day 6	API Testing
Day 7	Presentation

Version Control

- GitHub repo

Team Roles

- QA Lead
- Automation Engineer
- API Tester

Capstone Project Ideas (11 Use Cases)

1. E-Commerce Website Testing

- UI + API + BDD
- Login, cart, checkout

2. Banking Application Testing

- Transactions, balance, security validation

3. Healthcare Appointment System

- Booking, cancellation, API validation

4. Flight Booking

- Search flights, booking, payment APIs

5. Food Delivery App

- Order flow, restaurant APIs

6. Hotel Room Reservation

- Search Rooms, booking, payment APIs

7. Bus Ticket Reservation

- Search tickets, booking, payment APIs

8. Insurance Claim Processing System

- Claim submission, approval workflow

9. Inventory Management System

- Add/update/delete products, API sync

10. Ride Booking Application

- Book ride, driver allocation, fare calculation

11. Job Portal Application

- Profile creation, job apply, search filters

Capstone Project Evaluation Metrics (Total: 100 Marks)

S.No	Category	Sub-Criteria	Marks
1	Problem Definition & Test Strategy	Understanding of AUT (Application Under Test)	3
		Defined scope (UI + API + BDD)	3
		Test Strategy / Test Plan clarity	4
		Subtotal	10
2	Test Design & Scenario Coverage	Business scenarios identification	5
		Test cases (positive & negative)	5
		Edge cases & boundary conditions	5
		Subtotal	15
3	Framework Design & Architecture	Page Object Model (POM) implementation	5
		Modular & scalable design	5
		Reusability (utilities, configs, data handling)	5
		Subtotal	15
4	Selenium Automation (Python)	Locator strategy & element handling	5
		Synchronization (waits, stability)	5
		Code quality & maintainability	5
		Subtotal	15
5	BDD Implementation (Behave)	Feature files (Gherkin syntax)	4
		Step definitions mapping	3
		Hooks, tags & organization	3
		Subtotal	10
6	API Testing (Postman + Newman)	API collection design	4
		Assertions & validations	3

S.No	Category	Sub-Criteria	Marks
		Newman execution & CLI usage	3
		Subtotal	10
7	Test Execution & Reporting	Execution of UI + API + BDD tests	4
		Report generation (Allure/Extent/Newman)	3
		Results interpretation	3
		Subtotal	10
8	Defect Handling & Analysis	Defect identification	2
		Defect reporting (severity, steps)	2
		Root cause analysis	1
		Subtotal	5
9	Documentation & Presentation	Project report structure	4
		Framework documentation	3
		Final demo/presentation	3
		Subtotal	10
10	Version Control & Project Management	Git usage (commits, branching)	3
		Timeline & milestone tracking	2
		Subtotal	5
		TOTAL	100

Final Deliverables

- Selenium Automation Framework (Python)
- Behave BDD Feature Files
- Postman Collections + Newman Reports
- Test Report (PDF)
- GitHub Repository
- Demo Presentation